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Survey on Improving Efficiency of Edge Computing

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***Abstract* — The ongoing growth of Internet of Things (IoT) applications has resulted in an urgent need for computing solutions that are scalable and secure and can process information with the immediacy of real-time calculations. Edge computing can offer integrated local handling of data in accordance with all of these requirements. Cloud computing provides centralized processing as well as a degree of scalability, but lacks real-time data processing and local performance efficiencies. This paper presents an assertion of modifications that integrated frameworks, or hybrid domains of computing fog and mist computing, will alleviate the challenges of implementing edge or cloud computing technologies solely. The hybrid models offered in this paper can enhance performance, further increase privacy protection, and guarantee responsiveness in systems spread across essential sectors like healthcare, smart cities, and industrial systems.**

Keywords—EDGE Computing, Efficiency, Internet of Things, Resource Management

I. INTRODUCTION

The accelerations in digital transformation in sectors like healthcare, manufacturing and smart cities have fostered the large-scale development of IoT-based systems. These systems constantly generate plenty of data that requires timely processing and intelligent decisions to be made. While traditional (centralized) cloud computing has sufficient computing power, it deals with some drawbacks like latency limits and limitations imposed due to bandwidth availability or conditions that may lead to potential privacy concerns over well-established cloud computing services—specifically applications or scenarios that require action immediately.

Conversely, edge computing is the process of data and information processing from its origination point. Edge connects as close as possible to the source location, reducing latency to the extent of understanding and disputing communication time delays, while also minimizing bandwidth consumption. Moreover, it is important to highlight that edge devices lack any sort of processing capacity. The performance and privacy trade-offs between edge, cloud, and hybrid computing models are later compared in Figure 1 and Table 2 to highlight their suitability for real-time, scalable systems., new models that are being used to either create the appearance of edge or give edge processing systems another layer of processing that uses fog or mist computing in conjunction with edge computing rather than cloud computing. Such frameworks integrate edge and cloud processing, enhancing performance, privacy control and scalability of the system[1].

II. RELATED WORK

In recent years, considerable research has been dedicated to both edge and cloud computing. The media's attention in recent years has been heavily focused on edge computing versus cloud computing. Though cloud computing provides a centralized data repository with high processing capabilities that can scale to storage size, edge technology can eliminate latency and maximize data proximity[16]. There is a growing interest in hybrid solutions that incorporate the advantages of cloud and edge computing[2][3].

Fog computing provides an intermediary step between edge devices and the cloud that filters and preprocesses data locally[4]. Mist computing takes this process to the next level by applying computational intelligence on the edge node, allowing immediate responses to localized events. Further, federated learning methods and privacy-preserving algorithms are starting to help with confidentiality and security. Activity in load balancing, energy-efficient scheduling, and task offloading is helping to improve the effectiveness of beneficial distributed networks.

III. METHODOLOGY

The study uses a systematic review research design based on the PRISMA methodology to examine the progress being made in edge–cloud integration in IoT applications. The systematic review will search for academic literature published from 2022 to 2024 to investigate the emerging trends, challenges, and use cases. The study is structured around six discrete research questions that contextualize the literature selected for inclusion. In particular, the review used researcher developed techniques such as keyword co-occurrence maps, clusters of topics and visual ways to summarize data in order to extract and interpret findings concerning performance, scalability, privacy and architectural compromises[5].

IV. Analysis and Discussion

A. Performance Enhancement

Edge computing enhances application responsiveness by processing data as close to its origins as possible This performance benefit is quantitatively illustrated in Figure 1, where edge computing demonstrates the lowest latency, followed by hybrid systems., essential in real-time applications such as emergency response or autonomous systems. In a hybrid architecture, time-sensitive activity is computed locally and computationally heavy workloads are offloaded to the cloud[6][7][8].

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Recent developments, including intelligent task offloading based on reinforcement learning, have reduced energy consumption and network usage. Adaptively distributing workloads based on the current system situation has further enhanced responsiveness and resource utilization.

B. Improving Privacy and Security

Edge computing's local data management reinforces user privacy by limiting sensitive information transmission to outside servers. Data anonymity and selective sharing techniques reduce the likelihood of data leaks. Federated learning enables models to learn and optimize without losing the ownership of the data. [9][10].

In addition, blockchain technology combined with hybrid computing allows for secure and immutable proving and recording of transactions. This is necessary in data-sensitive industries such as healthcare, where ethical and legal compliance is of utmost concern.

C. Using Resources Wisely

As edge devices are typically limited in power and processing capacity, using resources wisely is important for better outcomes; this requires the intelligent management of resources. It is possible to use algorithms to analyze key variables such as energy to use, task difficulty, and the speed of response As seen in Table 2, edge computing offers minimal latency but limited scalability, underlining the importance of intelligent resource allocation in constrained devices.in relation to whether the task should be completed locally or offloaded[11] [12].

When utilizing mobile and aerial networks (for instance, drones), energy must be shared among communication, processing, and navigation. They can keep the system efficient with smart scheduling and low-power design approaches, while still ensuring accuracy and performance.

D. Real-time Applications: Healthcare

Healthcare is probably one of the biggest benefactors of hybrid computing Hybrid computing's balance between privacy and cost (as shown in Figure 1) makes it particularly suitable for sensitive domains such as healthcare [13], with devices like wearables embedded with sensors allowing continuous health monitoring, where the data can be analyzed in near-real-time for anomalies. At the same time, this allows using less bandwidth since the alarms do not constantly connect to the cloud. The responsiveness is also fast when anomalies are found[14][15].

Cloud-trained systems can provide improved diagnoses by fine-tuning general models with personalized data generated at the edge. These models not only benefit the patients, but they allow for remote medical assessments and ensure the protection of personal medical records.

V. Data Processing Models in Hybrid Systems

Hybrid systems allow for advanced data processing models that balance local processing and central processing. In more detail Hybrid systems offered support for:

• Federated Learning: allows organizations and edge devices to jointly and collaboratively train models together as a network without having to transfer raw data, maintaining data privacy[16].

• Distributed learning: divides a model's training work across multiple, devices, or nodes to improve efficiency and balance workloads.

• Decentralized Learning: Each device independently trains a complete model. This is typically done in disconnected environments or autonomous modes.

• Homomorphic encryption: provides privacy for computing over encrypted datasets.

VI. Comparative Analysis of Computing Architectures

TABLE I. Comparative Analysis of Computing Architectures

Aspect	Edge Computing	Cloud Computing	Hybrid (Fog/Mist)
Latency	Very Low	High	Low to Moderate
Scalability	Limited	Extensive	Moderate
Privacy	High	Moderate	High
Cost	Higher initial deployment	Lower entry cost	Balanced investment
Maintenance	Managed locally	Managed by service providers	Split responsibilities
Best Applications	Time-sensitive, localized tasks	Big data storage and analytics	Adaptive, real-time distributed systems

Hybrid architectures provide a balance between reflexivity at the edge and full computational strength in the cloud and retain the best characteristics of both models for contemporary IoT necessities[17].

VII. RESULT

This section compares the performance and operational efficiency of Edge, Cloud, and Hybrid (Fog/Mist) computing models across key parameters such as privacy, cost, latency, and scalability.

To begin with, a performance evaluation based on privacy and cost is shown in Table 1. As evident, Edge computing provides higher privacy scores but incurs greater costs due to localized deployment and infrastructure needs. In contrast, Cloud computing has lower associated costs but comparatively lower privacy levels. Hybrid models maintain a moderate balance in both aspects.

TABLE II. Operational Comparison: Privacy and Cost

Aspect	Computing Type	Value
Privacy	Edge Computing	2
Cost	Edge Computing	3
Privacy	Cloud Computing	1

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Cost	Cloud Computing	1
Privacy	Hybrid (Fog/Mist)	2
Cost	Hybrid (Fog/Mist)	2

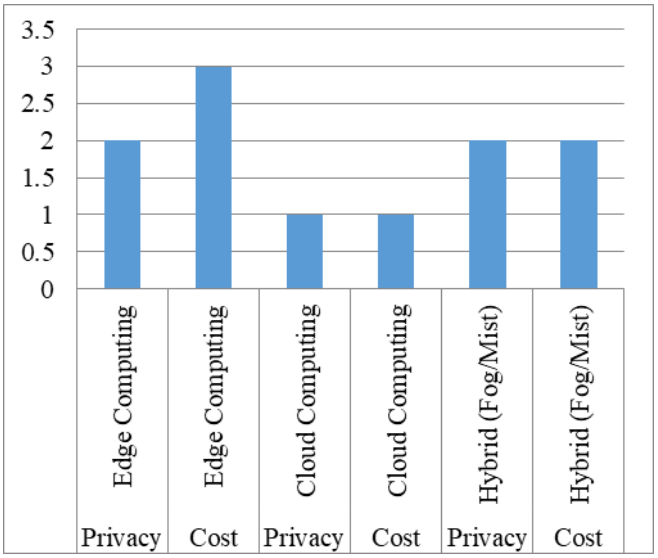


Fig. 1. Performance Comparison: Latency and Scalability

After evaluation, the operational parameters, specifically latency and scalability, which are summarized in Table 2. Edge computing shows minimal latency but has limited scalability. Cloud systems, on the other hand, scale extensively but are hindered by higher latency. Hybrid models offer a moderate balance in both areas.

TABLE II. OPERATIONAL COMPARISON: PRIVACY AND COST:

Aspect	Computing Type	Value
Latency	Edge Computing	1
Scalability	Edge Computing	1
Latency	Cloud Computing	3
Scalability	Cloud Computing	3
Latency	Hybrid (Fog/Mist)	2
Scalability	Hybrid (Fog/Mist)	2

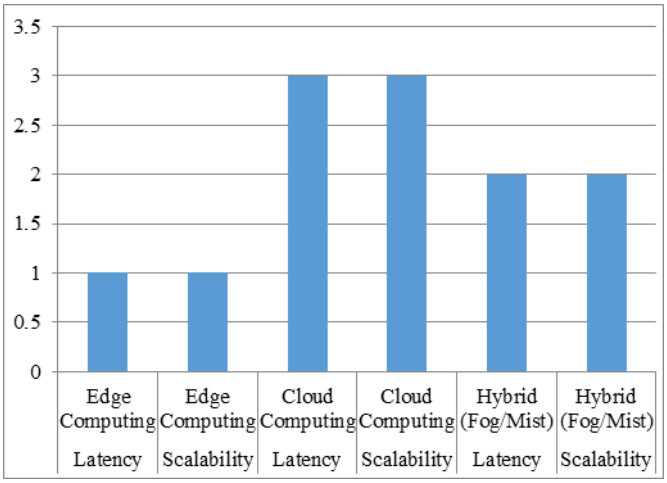


Fig. 2. Latency and Scalability Comparison of Edge, Cloud, and Hybrid Models

VIII. CONCLUSION

As IoT systems grow in usage there is more of a requirement for solutions that provide efficient, scalable, and privacy-aware data processing models. Edge provides real time response time and local control, while cloud provides a high resource computational platform. Hybrid models like fog and mist computing reduce latency, provide additional security, and make the best use of resources. In relation to future research, more emphasis should be placed on continuously lightweight AI models that can have a low impact running on edge devices, improved energy management solutions, and dynamic workload orchestration approaches, all of these areas of research will be vital in supporting the next generation intelligent applications such as intelligent healthcare, autonomous vehicular systems, smart manufacturing and emergency response systems.

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